Runtime comparison | Apple M2 / macOS

Historical paper timings from multiple recorded batches

Seconds | median of three runs

Problem	n	γ	**LSQN**	VI	PI	MPI-4	MPI-20	QPI-U	QPI-BT	QPI-S	iGMRES-PI	AA-I-S-5
Queue	512	0.99	0.0686	0.0913	**0.0095**	0.3627	0.0819	0.1231	0.1083	1.6564	0.1334	0.0557
		0.999	0.0807	0.9008	**0.0070**	3.7084	0.8271	0.6086	1.9123	12.2657	TO	F(0/3)
	1024	0.99	0.2730	0.1235	**0.0108**	0.3938	0.0924	0.1570	0.1416	8.1624	0.1648	0.0811
		0.999	0.4710	1.2390	**0.0073**	4.0788	0.9317	1.1310	2.6634	TO	TO	0.5668
	2048	0.99	0.5219	0.2096	**0.0131**	0.4503	0.1127	0.2128	0.1950	35.9137	0.1981	0.1866
		0.999	6.8800	1.9889	**0.0113**	4.6928	1.1415	2.0538	3.0101	TO	TO	1.2727
	4096	0.99	0.8099	0.3668	**0.0177**	0.5883	0.1448	0.3720	0.3487	TO	0.3105	0.2391
		0.999	26.9971	3.6819	**0.0154**	5.9282	1.4268	3.9456	3.7362	TO	TO	3.0200
Cluster	512	0.99	**0.0084**	0.3660	0.0086	0.2656	0.1229	0.1465	0.1420	0.0583	0.0114	0.0141
		0.999	**0.0090**	3.6804	0.0132	2.6734	1.2522	0.2718	0.2616	0.0623	0.0143	0.0285
	1024	0.99	**0.0255**	1.4226	0.0775	1.4510	0.5529	0.4859	0.4806	0.2918	0.0454	0.0339
		0.999	**0.0294**	14.1915	0.0738	14.1705	5.4892	0.8586	0.8876	0.4003	0.0514	0.0885
	2048	0.99	**0.0903**	5.5261	0.5014	4.6893	2.0326	1.9925	1.9293	0.8835	0.1715	0.1198
		0.999	**0.1059**	55.6762	0.5217	46.4943	19.9631	3.5692	3.5188	3.0827	0.1999	0.2670
	4096	0.99	**0.3225**	21.4234	4.0320	19.7953	8.4453	6.9703	6.9987	4.0106	1.2034	0.5047
		0.999	**0.4126**	TO	3.7347	TO	TO	13.8129	13.5234	4.6227	1.0582	0.7782
Garnet	512	0.99	0.0030	0.1031	0.0282	0.3897	0.0889	0.0026	**0.0023**	0.0367	0.0264	0.0061
		0.999	0.0032	1.0393	0.0279	3.9141	0.8929	0.0025	**0.0022**	0.0364	0.0126	0.0050
	1024	0.99	0.0040	0.1460	0.2276	0.4509	0.1045	0.0033	**0.0029**	0.1911	0.0120	0.0075
		0.999	0.0043	1.4631	0.2280	4.4497	1.0279	0.0033	**0.0030**	0.1915	0.0138	0.0070
	2048	0.99	0.0060	0.2643	1.7037	0.5586	0.1356	**0.0046**	0.0048	0.8312	0.0157	0.0164
		0.999	0.0066	2.6097	1.7375	5.4970	1.3149	0.0051	**0.0044**	0.8387	0.0168	0.0149
	4096	0.99	0.0101	0.4474	13.2972	0.7759	0.1887	0.0078	**0.0073**	3.4203	0.0180	0.0214
		0.999	0.0121	4.3800	13.3319	7.6461	1.8568	0.0084	**0.0072**	3.4289	0.0198	0.0224
Latent	512	0.99	0.0031	0.3763	0.0084	0.2760	0.1296	0.0017	**0.0017**	0.0089	0.0064	0.0025
		0.999	0.0028	3.8107	0.0104	2.6855	1.2602	0.0017	**0.0016**	0.0106	0.0067	0.0024
	1024	0.99	0.0079	1.4637	0.0546	1.4426	0.5730	0.0044	**0.0044**	0.0378	0.0264	0.0065
		0.999	0.0088	14.3604	0.0554	13.9815	5.4525	**0.0044**	0.0044	0.0359	0.0213	0.0073
	2048	0.99	0.0314	5.5946	0.3667	4.7967	2.0799	0.0163	**0.0159**	0.1491	0.0770	0.0239
		0.999	0.0288	F(2/3)	0.3658	50.9477	21.4017	0.0170	**0.0167**	0.1567	0.0730	0.0231
	4096	0.99	0.1404	21.6557	2.5239	17.8190	7.9514	**0.0645**	0.0666	0.7618	0.4790	0.0946
		0.999	0.1105	TO	2.9200	TO	TO	0.0636	**0.0626**	0.8865	0.3115	0.1071
Mean rank (lower is better)			3.5312	8.1875	4.8750	8.8438	6.4375	4.0625	3.5625	7.0156	5.0312	3.4531
Overall rank			2	9	5	10	7	4	3	8	6	1
Successful instances			32/32	29/32	32/32	30/32	30/32	32/32	32/32	28/32	28/32	31/32

Entries are median seconds from three runs. Bold marks the lowest successful time in each row, using unrounded values. Success requires all three runs to satisfy the independent residual check (infinity norm at most 1e-8). TO: all three runs timed out; F(s/3): only s runs succeeded. Per-run limits: 30 s for n at most 1024; 60 s otherwise. Ranks use unrounded times; unsuccessful methods tie below successful ones.

These historical paper values combine multiple timing batches; per-cell provenance is retained. Ranks are descriptive. These measurements are not an isolated CPU-performance comparison; see the archive metadata.